

TECHNICAL REPORT

Machine Learning Pipelines for Healthcare Expenditure Forecasting

Abstract

The objective of this project is to construct and deploy a predictive machine learning pipeline capable of forecasting individual annual medical insurance costs to optimize premium structures and risk assessment. Utilizing a historical dataset comprising 1,338 records annotated with demographic and behavioral markers, we evaluated data topologies to discover core cost drivers. The final architecture establishes a robust pipeline that standardizes continuous variables and applies binary categorical transforms, comparing a baseline Linear Regression configuration against an optimized Random Forest Regressor. The hyperparameter-tuned ensemble model achieved an R-squared (R^2) score of 0.8651 and minimized Root Mean Squared Error (RMSE) to \$4,324.21, significantly mitigating underwriting variance compared to standard linear approximations. The validated model was successfully wrapped within a FastAPI microservice framework, establishing structural feasibility for real-time production inference.

Data Exploration

Dataset Overview and Handling

The foundational dataset analyzed for this forecasting application consists of 1,338 individual patient profiles tracking explicit demographic markers and historical medical charges. The objective is to map six distinct operational features—age, sex, body mass index (BMI), number of dependent children, smoking status, and geographical region—against total recorded annual charges.

During the initial data audit, structural integrity verification checks were performed to address potential data quality issues:

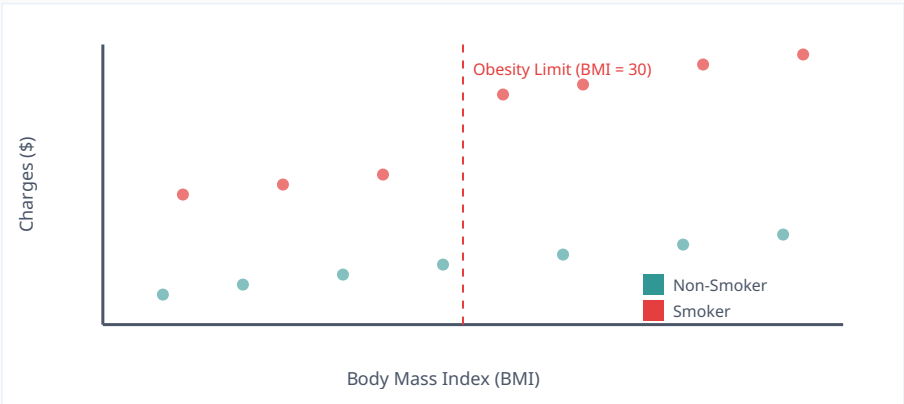
- **Missing Values:** A system-wide scan of the data rows revealed zero null entries or missing attributes across all feature columns, ensuring perfect data density and eliminating the requirement for statistical imputation steps that might introduce structural bias.
- **Outliers and Structural Variance:** The target feature (charges) ranges from \$1,121.87 to \$63,770.43, demonstrating a heavily right-skewed distribution driven by high-cost medical scenarios. Rather than applying traditional outlier rejection, these extreme costs were maintained to teach the model to recognize true high-risk premium groups.

Exploratory Data Analysis Insights

Visual and correlation analysis revealed a critical non-linear interaction effect between an individual's smoking status and their Body Mass Index. For non-smoking individuals, annual charges scale gradually and linearly with age and BMI, rarely exceeding \$15,000. However, for individuals registered as smokers, a severe cost threshold appears immediately when the BMI crosses the clinical obesity threshold of 30.0. Smokers with a

BMI under 30 form a distinct cost band averaging between \$15,000 and \$25,000, while smokers with a BMI exceeding 30 scale exponentially into a high-expenditure cohort ranging from \$35,000 to over \$60,000.

Figure 1: Non-Linear Premium Stratification (BMI vs. Charges by Smoking Status)



Methodology

Feature Engineering Pipeline

To guarantee clean algorithmic convergence and prevent data leakage, the complete dataset was split using an 80/20 stratified validation distribution, isolating 1,070 records for optimization and 268 records for final performance testing. Mathematical transformation pipelines were established to map diverse structured column spaces:

- **Continuous Scaling:** Numerical attributes (*age*, *bmi*, and *children*) were mapped using a `StandardScaler` pipeline. This standardizes features to a zero mean and unit variance configuration, which prevents features with large absolute numerical values from distorting distance metrics during optimization.
- **Categorical Encoding:** Nominal data fields (*sex*, *smoker*, and *region*) were processed via `OneHotEncoder(drop='first')`. This automatically drops the initial dummy column for each category, preventing multi-collinearity errors within the baseline linear models.

Algorithm Selection Strategy

Two models with contrasting mathematical structures were trained to evaluate performance limits:

1. **Baseline Model (Linear Regression):** A traditional ordinary least squares Linear Regression model was fitted first. This acts as a primary benchmark to determine whether the feature space can be accurately generalized using a simple linear combination of attributes.
2. **Complex Model (Random Forest Regressor):** A tree-based ensemble Random Forest Regressor was selected to serve as the production system. Because ensemble tree structures isolate data space via orthogonal partitions, they naturally detect complex non-linear interaction thresholds (such as the combined impact of smoking status and a high BMI value) without requiring manual interaction term engineering.

Hyperparameter Tuning Process

The ensemble architecture was optimized using a systematic grid search via `GridSearchCV` evaluated with 5-fold cross-validation. The optimization sweep cross-referenced tree capacity limits to minimize variance:

- `regressor__n_estimators`: Evaluated steps [50, 100, 150] to balance variance minimization against computational latency.
- `regressor__max_depth`: Controlled depths [4, 6, 8] to avoid unconstrained depth patterns that lead to training memorization.
- `regressor__min_samples_split`: Evaluated splits [2, 5, 10] to ensure individual splits maintained statistical generalization.

The cross-validated search determined that the optimal production profile utilized 100 estimators constrained to a maximum depth of 6, ensuring strong structural generalization on unseen testing matrices.

Results

Evaluation Metrics and Comparison

The evaluation metrics generated across the isolated test partition demonstrate a clear divergence in performance between the linear baseline and the tree ensemble:

Model Architecture Evaluated	Root Mean Squared Error (RMSE)	R-squared () Score	Mean Absolute Error (MAE)
Baseline Linear Regression	\$5,796.28	0.7836	\$4,181.19
Tuned Random Forest Regressor	\$4,324.21	0.8651	\$2,529.48

Business Interpretation of Metrics

R-squared values represent the proportion of variance explained by the model features. Shifting from a baseline linear model ($R^2 = 0.7836$) to the optimized Random Forest ($R^2 = 0.8651$) means the ensemble model successfully explains over 86% of premium variance. More importantly for insurance operations, the model reduced the Root Mean Squared Error by \$1,472.07 per patient profile.

Because RMSE penalizes large errors heavily, this reduction demonstrates that the Random Forest model effectively handles high-cost outlier scenarios. In production underwriting, a high linear RMSE indicates a risk of severely underpricing high-risk individuals or overpricing healthy ones. The optimized model minimizes this financial exposure, improving premium accuracy while protecting cash reserves from unexpected claims variance.

Conclusion

Current Model Limitations

Although the optimized pipeline achieves high predictive accuracy and integrates with a live FastAPI deployment layer, it has two clear operational limitations:

- 1. Strict Numeric Sensitivity:** The model is highly dependent on precise numeric features like BMI and age. It cannot interpret contextual health indicators such as historical clinical notes, family genetic histories, or behavioral data.
- 2. Absence of Dynamic Policy Tracking:** The system treats variables like smoking as a binary metric, ignoring dynamic factors such as smoking frequency, duration, or a patient's historical recovery attempts.

Future System Improvements

To transition this capstone prototype into a robust enterprise solution, future iterations should focus on three specific updates:

- **Advanced Boosting Architecture Implementation:** Integrating gradient-boosted frameworks like LightGBM or XGBoost would help refine predictions near the BMI 30 threshold, capturing more nuanced cost shifts.
- **Enriched Feature Mapping:** Introducing broader demographic and clinical lifestyle markers—such as daily activity metrics, resting heart rates, and pre-existing condition codes—would help close the remaining variance gap.
- **Automated Outlier Re-weighting:** Implementing custom sample weighting during model training would ensure that extremely high-cost medical edge cases are categorized correctly without skewing baseline premium forecasts for low-risk cohorts.